

ULIP Integration Requirement Document

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Revision History

Version	Date	Author	Reviewer	Significant Changes
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2.0	10/03/2023	Ayush Arya	Nayab Ali	Added CFS ICD integration API
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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for LDB system from ULIP system.

This document details the following points:

1. **Data Integration between LDB and ULIP web service:** Identification of operational points where data integration between LDB and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

User will share information such as **container number** based on data send by user
ULIP system will call respective **LDB** API for accessing information.

Following are the points for accessing container information: -

1. This API will share data based on information provided by user.
2. ULIP will expose API for getting data from LDB API.
3. **LDB/01** API will take **container Number** from user and get data from LDB API.
4. **CFSICD/01** API will take the **port Code, destination Code, from and to**, input parameter from the user and get the data from LDB API.

1.2 Data Integration between LDB System and ULIP application

User will share information to ULIP system based on data shared ULIP will provide data to user.

1.3 LDB/01

This API will take **container number** and connect with LDB API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	containerNumber	container number as available in LDB system.	^[a-zA-Z0-9]{8,11}\$	Required	Min 8 Max 11

1.3.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **container number** for which detail is required.
3. LDB API will share details of container tracking information which exist in LDB system in JSON format.
4. LDB Integration will add response from LDB system as value to **response** key in response body.
5. ULIP system will provide the data in JSON Object format in response body.

Example:

Container number **ACMP1000010** is shared by ULIP system, then LDB-integration will make a request to LDB API.

Request:

The request must be of the following format, where container number is passed within the URL body in JSON format.

For Production:

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/LDB/01>

```
{"containerNumber": "ACMP1000010"}
```

Curl URL

```
curl --location --request POST
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/LDB/01' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsIm1hdCI6MTYyOTI2ODM4NSwiYXBwcyI6ImRh
dGFwdXNoIn0.x3REIfQ7GGF4-
ysUexOBegV_ON7Ex4RitT62ipKnAI_RyMEfQgdV_DBzAJnz118yoieTAqsRyIozj-pAwudB_w' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data-raw '{"containerNumber": "ACMP1000010"}'
```

For Staging:

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/LDB/01>

```
{"containerNumber": "NSST1234570"}
```

Curl URL

```
curl --location --request POST
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/LDB/01' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsIm1hdCI6MTYyOTI2ODM4NSwiYXBwcyI6ImRh
dGFwdXNoIn0.x3REIfQ7GGF4-
ysUex0BegV_ON7Ex4RitT62ipKnAI_RyMEfQgdV_DBzAJnz118yoieTAqsRyIozj-pAwudB_w' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data-raw '{"containerNumber": "NSST1234570"}'
```

For obtaining authentication token, refer [Section 1.5](#)

Response

To acknowledge the above request, ULIP will send the below response -

a) In case of invalid format

1. Invalid container number

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered
at: containerNumber. Format should follow [A-Z]{4}[0-
9]{4,7}"
}
```

b) In case of container number do not exist in LDB system

```
{
  "response": [
    {
      "responseStatus": "FAILURE",
      "message": {
        "text": "Container details not found for the given container number: TCLU8538800"
      }
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

c) In case container number exists

```
{
  "response": [
    {
      "response": {
        "eximContainerTrail": {
          "cntrDetail": {
            "cntrno": "NSST1234570",
            "refflg": "Yes",
            "cntrsize": 40,
            "isocode": "22G3",
            "containertype": null
          },
          "dpd_dpe": {
            "c_import": {
              "dpd": "N",
              "destination": null
            },
            "c_export": null
          },
          "last_event": [
            {
              "serialno": 1,
              "eventname": "PORT OUT",
              "currentlocation": "Raigad/Nhava Sheva Freeport Terminal (NSFT)",

```

```
11:00:09",
    "division": null,
    "timestamptimezone": "2023-03-29
    "timezoneabvr": "IST",
    "latitude": 18.950149,
    "longitude": 72.95123,
    "containernumber": "NSST1234570",
    "timeinms": 1680067809000,
    "transportmode": "TRUCK",
    "type": "I",
    "isempty": "Y"
  }
],
"res_Message": "Success",
"trackLog": [
  {
    "serialno": 1,
    "eventname": "PORT OUT",
    "currentlocation": "Raigad/Nhava
Sheva Freeport Terminal (NSFT)",
    "division": null,
    "timestamptimezone": "2023-03-29
11:00:09",
    "timezoneabvr": "IST",
    "latitude": 18.950149,
    "longitude": 72.95123,
    "containernumber": "NSST1234570",
    "timeinms": 1680067809000,
    "transportmode": "TRUCK",
    "type": "I",
    "isempty": "Y"
  },
  {
    "serialno": 2,
    "eventname": "PORT IN",
    "currentlocation": "Raigad/Nhava
Sheva Freeport Terminal (NSFT)",
    "division": null,
    "timestamptimezone": "2023-03-28
09:00:08",
    "timezoneabvr": "IST",
    "latitude": 18.950149,
    "longitude": 72.95123,
    "containernumber": "NSST1234570",
```



```

        "timeinms": 1679974208000,
        "transportmode": "VESSEL",
        "type": "I",
        "isempty": "Y"
    },
    ],
    "vessel_ata": null,
    "vessel_atd": null,
    "vessel_eta": [
        {
            "eventid": 24,
            "infotime": 1680659629000,
            "eventname": "EXPECTED VESSEL
ARRIVAL",
            "orgname": "Nhava Sheva Freeport
Terminal (NSFT)",
            "timeinms": 1679978002000,
            "timestimestamp": "2023-03-28
10:03:22",
            "latitude": null,
            "longitude": null,
            "vesselname": "APR EXPRESSS",
            "vesselimo": null,
            "cntrcycleid": 1,
            "shippingline": "TRANSPORT
CORPORATION OF INDIA LTD"
        }
    ],
    "vessel_etd": null,
    "vessel_gate_cutoff": null
},
"domesticContainerTrail": {
    "cntrDetail": null,
    "dpd_dpe": null,
    "last_event": null,
    "res_Message": "NO RECORD FOUND",
    "trackLog": null,
    "vessel_ata": null,
    "vessel_atd": null,
    "vessel_eta": null,
    "vessel_etd": null,
    "vessel_gate_cutoff": null
}
},

```

```

        "responseStatus": "SUCCESS",
        "message": null
    },
    "error": "false",
    "code": "200",
    "message": "Success"
}

```

Status Code 200: container data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: Server is not responding.

1.4 CFSICD/01

This API will take “**port Code, destination Code, from and to**” and connect with LDB API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	portCode	Port Code as available in LDB system.	[a-zA-Z0-9]{12}	Required	Max and Min 12
2.	destinationCode	Destination Code as available in LDB system.	[a-zA-Z0-9]{1,20}	Required	Min 1 and Max 20
3.	from	From Date as available in LDB system.	^([0-2][1-9] [1-3][01])([0][1-9] 1[0-2])\d{4}\$	Required	Max and Min 8
4.	to	To Date as available in LDB system.	^([0-2][1-9] [1-3][01])([0][1-9] 1[0-2])\d{4}\$	Required	Max and Min 8

1.4.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **port Code, destination Code, from and to** for which detail is required.
3. CFS ICD API will share details of “**port code, destination code, from and to**” information which exist in LDB system in JSON format.
4. CFS ICD Integration will add response from LDB system as value to **response** key in response body.
5. ULIP system will provide the data in JSON Object format in response body.

Request:

The request must be of the following format, where port code, destination code, from and to details is passed within the URL body in JSON format.

For Staging:

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/CFSICD/01>

```
{
  "portCode": "ORG000004818",
  "destinationCode": "INTUT1HAR1",
  "from": "20012023",
  "to": "29012023"
}
```

Curl URL

```
curl --location --
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/CFSICD/01' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlOGIwIiwiaWF0IjoxNjc4NDQ2ODYyLCJhcHBzIjoizGF0YXB1c2gifQ.GnWvsv9FcQlfxPMYCZiP2-yivi0oJi2P5ybS9XCyPLtN9IvL3WCB-6upIIMlXFG6lTD_ct3u6Nehux5TQQ4NQ' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{
  "portCode": "ORG000004818",
  "destinationCode": "INTUT1HAR1",
  "from": "20012023",
  "to": "29012023"
}'
```

For Production:

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/CFSICD/01>

```
{
  "portCode": "ORG000004818",
  "destinationCode": "INTUT1HAR1",
  "from": "20012023",
  "to": "29012023"
}
```

Curl URL

```
curl --location --request POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/CFSICD/01' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlOGIwIiwiaWF0IjoxNjc4NDQ2ODYyLCJhcHBzIjoizGF0YXB1c2gifQ.GnWvfsv9FcQlfxPMYCZiP2-yivi0oJi2P5ybs9XCyPLtN9IvL3WCB-6upIlMlXFG6lTD_ct3u6Nehux5TQQ4NQ' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{
  "portCode": "ORG000004818",
  "destinationCode": "INTUT1HAR1",
  "from": "20012023",
  "to": "29012023"
}'
```

For obtaining authentication token, refer [Section 1.5](#)

Response

To acknowledge the above request, ULIP will send the below response -

a) In case of invalid format

1. Invalid portCode number (or any field)

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: portCode. Format should follow [a-zA-Z0-9]{12}"
}
```

b) In case of any field missing in request body

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "to:is mandatory field."
}
```

c) In case port Code, destination Code, from and to exist

```
"response": [
  {
    "response": {
      "message": {
        "text": "COMMON_002"
      },
      "objectType": "Optional",
      "object": [
        {
          "containerNo": "TCNU4866067",
          "containerSize": "40",
          "category": "I",
          "entryTime": "01-01-2023 01:06:18",
          "entryMode": null,
          "exitMode": "T",
          "exitTime": "02-01-2023 09:16:39",
          "isRefrigerated": "N",
          "wagonNo": null,
          "loadingTime": "1672630836000",
          "destination": "INTUT1HAR1",
          "vesselName": "FSL KELANG",
          "isoCode": "4500",
          "isEmpty": "Y",
          "containerType": null
        },
        {
          "containerNo": "TEMU6678583",
          "containerSize": "40",
          "category": "I",
          "entryTime": "01-01-2023 05:27:14",
          "entryMode": null,
          "exitMode": "T",
```

```
        "exitTime": "02-01-2023 09:18:44",
        "isRefrigerated": "N",
        "wagonNo": null,
        "loadingTime": "1672629801000",
        "destination": "INTUT1HAR1",
        "vesselName": "FSL KELANG",
        "isoCode": "4500",
        "isEmpty": "Y",
        "containerType": null
      }
    ],
    "responseStatus": "SUCCESS"
  }
},
"error": "false",
"code": "200",
"message": "Success"
}
```

d) In case from date is greater than to date

```
{
  "response": [
    {
      "response": {
        "date": "2024-12-06T05:11:10.080+00:00",
        "message": "From date cannot be greater than to date.",
        "description": "uri=/container/cfslcdData"
      },
      "responseStatus": "ERROR"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

Status Code 200: Data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: Server is not responding.

1.5 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in LDB system through a data exchange URL that has following form:

For Production:

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/LDB/01>

Body:

```
{"containerNumber": "ACMP1000014"}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until token will not get expired.

Brief summary of HTTP Basic Authentication is as follows:

First time when User will hit ULIP secured API through username and password for access token as follow-

```
curl -X POST
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login'
-H 'accept: application/json'
-H 'content-type: application/json'
-data '{"username": "xxxxxx", "password": "XXxxx@12345"}'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJpbnRlcmNvbnQlLCJpYXQiOiJlODU1NjYyMzcsImFwcHMiOiJkYXRhcHVzaCJ9.jb-23Dbkwpz-3ptsuuZofzH-ce7jd0xR5WLkFfBP0_Dzjop01dSx90amqsjsqUAryLUPtUDiMX07oHG4L2AffQ
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for container details as follow -

curl --Location --request POST

```
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/LDB/01' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsImIhdCI6MTYyOTI4MzQxMiwiYXBwcyI6ImRh  
dGFwdXNoIn0.2fanfJuecjmwyT0Z0-H3qLpA4bKEnurPxx5yFeV_pw8h5TbLs5sBQFXu1Gq-  
FF5GM0fZV2a7EAOPvazmJC1QTw' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data-raw '{"containerNumber": "ACMP1000014"}'
```

For Staging:**curl -X POST**

```
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login  
-H 'accept: application/json'  
-H 'content-type: application/json'  
-data '{"username": "xxxxxxx", "password": "XXxxx@12345"}'
```

Authorization:**Bearer**

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJpbmRlcmNvbnQiLCJpYXQiOiJlODU1NjYyMzcsImFwcHMh  
OiJkYXRhcHVsZaCJ9.jb-23Dbkwpz-3ptsuuZofzH-  
ce7jd0xR5WlkFfBP0_Dzjop01dSx90amqsjsuAryLUPtUDiMX07oHG4L2AfFQ
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for container details as follow -

curl --Location --request POST

```
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/LDB/01' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsImIhdCI6MTYyOTI4MzQxMiwiYXBwcyI6ImRh  
dGFwdXNoIn0.2fanfJuecjmwyT0Z0-H3qLpA4bKEnurPxx5yFeV_pw8h5TbLs5sBQFXu1Gq-  
FF5GM0fZV2a7EAOPvazmJC1QTw' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data-raw '{"containerNumber": "NSST1234570"}'
```